DATA ITEM DESCRIPTION

Title: VERIFICATION TEST REPORT

Number: DI-NUSU-82342
AMSC Number: 10237
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20210331
Limitation: N/A
GIDEP Applicable: No
Project Number: NUSU-2021-004

Use/Relationship:

The Verification Test Report applies to both parts of military standard MIL-STD-188-125, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Ground-Based Facilities Performing Critical, Time-Urgent Missions*. This DID applies to fixed facilities in accordance with Part I (MIL-STD-188-125-1A) and to transportable systems in accordance with Part II (MIL-STD-188-125-2). This DID contains the format, content, and intended-use information for the data deliverable resulting from the verification-test tasks performed in accordance with section 4.4.3 and Appendices B and C of each part of MIL-STD-188-125. This DID also identifies the same categories of information required for the threat-level illumination verification-test tasks described in Part II's Appendix D, as applicable.

The purpose of verification testing the HEMP-protection subsystem (HPS) is to demonstrate that the installed protection provides the required HEMP hardness. These tests are performed after the construction and acceptance testing are complete and after the equipment is installed and functioning to determine if the operational system suffers mission-aborting damage or upset due to simulated HEMP excitations. Verification is normally a Government-conducted test and is not part of a facility construction contract.

The Verification Test Report documents the test's results of the HPS' performance, and also describes the test-execution, data-recording, analysis, inspection, and reporting procedures used by the contractor to assess and document these results. This report includes a discussion of recommended corrective actions, if needed. The report provides the means for the Government to evaluate and certify the HPS' performance results, in that the report demonstrates the HPS' compliance with respect to MIL-STD-188-125's verification-test pass and fail requirements. The Department of Defense's HEMP-certification authority requires that a documented verification test report be submitted as part of a HEMP-certification request package for a facility or system.

If a portion of the HPS' components has already been tested by a previous test program, then documentation of those previous tests, including reports and test data, is attached as an appendix.

Final data from the report will be applied to subsequent re-verification planning and testing efforts, and will be applied to data item description DI-NUSU-8 2 3 4 3 – MIL-STD-188-125 Hardness Maintenance (HM) and Hardness Surveillance (HS) Plan.

DISTRIBUTION STATEMENT A. Approved for public release.

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Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The Verification Test Report's classification shall be in accordance with any relevant facility- or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities. (DTRA's security classification guide is available by mail request to ATTN: RD-NTSA, Dr. William Linzey, PhD, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA 22060-6201.)

2. Format. Contractor format is acceptable except for the electronic data file format that is specified in sections 3.3, 3.3.2, and 3.3.3 below.

3. Content. The Verification Test Report shall document the test-execution, data-recording, analysis, inspection, and reporting procedures used to comply with MIL-STD-188-125's section 5.13 and Appendices B and C of Parts I and II, and also Appendix D of Part II, including:

a. Pulsed current injection (PCI) test procedures per Appendix B

b. Continuous wave immersion (CWI) test procedures per Appendix C

c. Threat-level illumination (TLI) test procedures per Appendix D (MIL-STD-188-125-2)

In addition to the written test report, the data obtained from the testing shall be presented electronically in comma separated values (CSV) format. The test report shall document instrumentation, calibration, processing procedures associated to this data. The electronic data format and guidance for users to locate specific data within this format shall be described in the written test report. A short summary of key facts and assumptions in the main text, with large amounts of supporting data in appendices, is encouraged.

3.1 Administrative data. The Verification Test Report shall contain an administrative section covering the following:

a. Contract number, CDRL number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the test report

3.2 Summary information. The Verification Test Report shall summarize the following:

3.2.1 Introduction, to include:

a. General objectives of the verification test

b. A general description of the HPS' topology that was tested

c. A statement of any assumptions or limitations associated with the verification testing

d. A summary of the verification test's pass and fail requirements per MIL-STD-188-125

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e. A specific statement of whether or not the HPS passes all requirements, and a summary listing of any requirements that were not passed

3.2.2 Scope. A general description of the overall test-requirements matrix that was used to verify the HPS' compliance with MIL-STD-188-125's performance requirements, and an associated listing of the test results. Also, a description of the relative roles of tests, inspections, and analyses shall be provided.

The following topics shall be included:

a. Synopsis of the PCI, CWI, and TLI test procedures performed (as applicable), and references to detailed procedures

b. Pass or failure performance results with respect to all requirements

c. Anticipated impact of any failures

d. Recommendations to resolve any failures

e. Lessons learned for the HPS' hardness and testability

3.2.3 Methods of verification. A general description of the procedures that were used to verify the HPS' compliance with each verification test performance requirement as listed in section 3.3 below.

3.2.4 Engineering factors. Any important engineering factors affecting the verification-test procedures, such as: physical-accessibility constraints of the facility's or system's topology, resource limitations, safety and security protocols, and precedent reports.

3.3 Detailed information. The overall Verification Test Report shall include the specific results of the SE and PCI testing, as well as describe in detail the test procedures and the associated test equipment. The Verification Test Report shall be prepared, as applicable: in three portions, one each for the PCI and CWI testing (as detailed in Appendices B and C of both Part I and Part II of MIL-STD-188-125) and one portion, where applicable, for the TLI testing (as detailed in Part II's Appendix D). These test-report portions shall be detailed in three separate sections of the overall Verification Test Report.

For the verification test of a transportable ground-based system, the test report shall note whether TLI tests were performed in lieu of CWI testing, or vice versa, and include an explanation for why this decision was made.

3.3.1 The overall test report. The main portion of the test report shall contain the following detailed information:

a. An identification of the facility or system that was tested; including photographs, figures, or diagrams

b. The facility's location, or the location of the system during its testing

c. Identification of the test-team personnel and of the Government witnesses that were present

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d. A description of the HPS' topology, to include: drawings showing the locations of all points of entry (POEs) and POE protective devices (PPDs) of each low-risk electromagnetic (EM) barrier, equipment layouts inside each low-risk EM barrier; the installation of mission critical systems (MCS) outside of each low-risk EM barrier and their associated special protective measures (SPMs); cable interconnections external to each low-risk EM barrier; the presence of any long line protection modules (LLPMs) associated with transportable ground-based systems; and a description of each HEMP-protection measure

e. Descriptions of the data acquisition equipment, to include: the manufacturer, model, and serial numbers of each equipment item; detailed calibration procedures for the equipment; calibration-traceability documentation; and a layout showing the interconnectivity of the overall data acquisition instrumentation system

f. Descriptions of the test procedures, including: system, subsystem, and circuit configuration requirements; equipment operating states, test configuration diagrams; test point locations and descriptions; and functional monitoring provisions

g. Descriptions of methods for measuring noise (instrument inherent and local or ambient)

h. Descriptions of the models, techniques, and tools used for preliminary analyses and predictions, and their specific application to the specific facility or system that was tested

i. Descriptions of the methods that were applied to quantify and evaluate any failures of the HPS' performance with respect to all pass and fail criteria

j. Identification of the pass and fail criteria, and of any deviations from the requirements of MIL-STD-188-125 that occurred, along with an explanation of why these deviations were necessary

k. Descriptions of the data-management procedures that were used, to include: data quality control procedures, data acceptability criteria, annotation and preservation of data records, and the methods used to compare results to pass and fail criteria

l. Identification of any safety risks encountered, and descriptions of procedures that were used to mitigate these risks, including any electromagnetic radiation hazards and electrical shock hazards

m. Descriptions of the data-security and physical-security procedures that were applied

n. A description of the test schedule as it was executed, including a listing of the prioritization of measurements

o. A description of the methods that were used to simulate the operational performance of mission critical systems (MCS) during the testing if the actual MCS operation during the testing was impractical

p. A written confirmation that the required MIL-STD-461 testing of the first articles of all MCS installed outside the shield was completed with results available prior to the verification test

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q. Descriptions of how the raw and processed test data were recorded, and the electronic data format used to store the data

r. Summary tables of measured quantities, including tables of PCI residuals, and CWI and TLI results, with calculations and comparisons to pass and fail criteria, and depicting any associated onsets of MCS upset or damage

s. Copies of measured test results, including all raw and processed test oscillographic waveform test data in engineering units, with a description of the electronic data format and guidance for users to locate specific data within this format

t. Findings from investigations into causes of failures, if any; and corrective actions recommended or, if performed, their re-test results

u. A discussion of the uncertainties involved with the test procedure and measured quantities and any issues regarding the reproducibility of test results

v. Descriptions of lessons learned, to include any insights learned during testing that could be applied to improving HPS performance and improving testing methodologies

w. If applicable, proposed improvements to the Hardness Maintenance and Hardness Surveillance plans (data item description DI-NUSU-####)

x. Test conclusions, including a definitive statement of the hardness of the HPS, based upon the results of the complete battery of verification tests and the supporting analyses

y. References to applicable test plans and procedures

3.3.2 The PCI portion of the test report. This portion of the Verification Test Report shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125 Appendix B's section B.4.7 and Tables B-II and B-III

b. Identification, with diagrams, of the locations of each HPS component that was subjected to PCI injection and an explanation for why these specific points were selected

c. Identification of the transfer impedance of the current-measurement probes and measurement range of the end-to-end data acquisition system

d. A table listing the PCI drive levels and associated residual-current measurements per test point, thereby identifying the PCI measurement's compliance (or failure) to meet the maximum allowable residual norm attribute response characteristics

e. Measured raw data for each test point in the form of data plots and tables, along with sensor calibrations and instrumentation settings required to convert the data to engineering units

f. The data obtained from the testing shall be presented electronically in CSV format; the electronic data format and guidance for users to locate specific data within this format shall be described in the written test report

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g. Descriptions of the methods that were used to determine the onset of any MCS upset or damage, and the locations for which observations were made

h. Identification of any special requirements of the facility operations that impacted HEMP testing

3.3.3 The CWI portion of the test report. This portion of the test report, if applicable, shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125 Appendix C's section C.4.4

b. Descriptions of the CWI antenna's field map characteristics, and the methods that were used to monitor the incident field and its relationship to these field map characteristics

c. Identification, with diagrams, of the locations and orientations of the facility or system under test with respect to the CWI transmit antenna's and receive antenna's position and polarization, and the placement of other test elements with respect to the CWI antenna (e.g., MCS outside the low-risk EM barrier), which were required to achieve functionality

d. A table listing all test point locations

e. Identification of the following instrumentation locations, with an explanation for why these specific points were selected:

   i. Electric and magnetic field sensors

   ii. Reference sensor

   iii. Representative MCS cable measurement test points both within, and external to, all electromagnetic barriers

   iv. Other measurement points as required

f. Identification of the measurement-range sensitivities of the field probes, current probes, and data acquisition system

g. Measured raw data for each test point in the form of data plots or tables, along with sensor calibrations and instrumentation settings required to convert the data to engineering units as needed, shall be provided; also, a copy of all final quality control (QC) processed data shall be provided

h. The data obtained from the testing shall be presented electronically in CSV format; the electronic data format and guidance for users to locate specific data within this format shall be described in the written test report

i. A description of the algorithms that were used to extrapolate the CWI results to the HEMP-threat level environments

j. A description of any alternative test methods that were used for verifying an electromagnetic barrier's performance if a CWI test was not possible due to constraints posed by the site's or system's location or construction

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3.3.4 The TLI portion of the test report. This portion of the Verification Test Report, if applicable, shall contain the following detailed information:

a. Detailed test procedures in accordance with Appendix D's section D.4 of MIL-STD-188-125-2

b. Descriptions of the TLI antenna's field map characteristics, and the methods used to monitor the incident field and its relationship to these field map characteristics

c. Identification, with diagrams, of the placement of subsystems and equipment under test in the TLI simulator's working volume and placement of other system elements, which were required to achieve functionality, with respect to the simulator working volume.

d. Diagrams of the interconnecting cable layouts

e. A table listing all test point locations

f. Identification of the following instrumentation locations, with an explanation for why these specific points were selected:

   i. Electric and magnetic field sensors

   ii. Reference sensor

   iii. Representative MCS cable measurement test points both within and external to all electromagnetic barriers

   iv. Other measurement points as required

g. Measured raw data for each test point in the form of data plots or tables, along with sensor calibrations and instrumentation settings required to convert the data to engineering units as needed, shall be provided

h. The data obtained from the testing shall be presented electronically in CSV format; the electronic data format and guidance for users to locate specific data within this format shall be described in the written test report

i. Identification of the measurement-range sensitivities of the field probes, current probes, and data acquisition system

3.4 Other information sources. When other available information sources contain data required by the Verification Test Report, these sources shall be referenced, rather than duplicated, within the test report.

End of DI-NUSU-82342

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